CS 142: Program design and methodology
Spring Term, 2025

Java review

Classes

Here is the format for a class, in general (left column) and in a specific example (right column).

```
public class className {

    attributes

    constructors

    other methods

}
```

```
public class SillyClass {
    private int value;

    public SillyClass(int value) {
        this.value = value;
    }

    public int getValue() {
        return value;
    }

    public void setValue(int newValue) {
        value = newValue;
    }
}
```

The code for a class goes into the file named after the class; the example code `SillyClass` should go into `SillyClass.java`.

Your program will need a special method called `main`, which is where it will start executing. The template for such a method is

```
public static void main(String[] args) {
}
```

Variables

Here are the main variable types and their main operations:

1. `int`:+, −, *, / (integer division; rounds down), % (mod operator; gives remainder)
2. `double`: +, −, *, /
3. `boolean`: &&, ||, !
   To get a `boolean` from a numeric type, use ==, ! =, <, >, <=, or >=.
4. `String`: `.length()`, `.equals()`, `.substring()`

Variables must be declared as attributes, local variables, or method parameters before they can be used. To create objects, you need to both declare and create the object using `new` as in the following:

```
SillyClass theObj;              //declares theObj as a reference to a SillyClass object
theObj = new SillyClass(42);    //creates object for theObj to refer to
                                //calls its constructor
```

Arrays

A special kind of object is an array, which stores a group of variables with the same type in different numbered cells. The type of an array is the type of thing it stores followed by [], as in `int[]`. These must be declared and then created with `new` giving the size of the array:

```
int[] nums;           //declares nums to be a reference to an  array of ints
nums = new int[100];  //creates an array containing 100 ints
```

Arrays of objects require a call to `new` to create the array and then a call to new for each object stored in it:

```
SillyClass[] bunchOfSillies;
bunchOfSillies = new SillyClass[3];       //creates array, but not objects in it
bunchOfSillies[0] = new SillyClass(19);
bunchOfSillies[1] = new SillyClass(-8);
bunchOfSillies[2] = new SillyClass(65);
```

Conditionals (If statements)

if(*booleanExpression*) {
    thenClause
}

```
if(x < 28) {
    System.out.println("It's small");
}
```

Or...

if(*booleanExpression*) {
    thenClause
} else {
    elseClause
}

```
if(x == 13) {
    System.out.println("Unlucky!");
} else {
    System.out.println("You're fine");
}
```

Loops

While loops:

`while`(*test*) {
    body
}

```
while(n != 1) {
    if(n % 2 == 0) {
        n = n/2;
    } else {
        n = 3*n + 1;
    }
}
```

For loops:

`for`(*initialization*; *test*; *increment*) {
    body
}

```
for(int i=0; i<10; i++) {
    System.out.println(i+1);
}
```